Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

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***Abstract*—Location-based social networks (LBSNs) record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can act ahead of demand. Two coarse questions usually suffice: the next category (the kind of place) and the next region (which part of the city). These are normally handled by separate models, so we ask whether one model can learn both, and what sharing a single representation costs. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across five U.S. states, this lifts next-category prediction by a wide margin over a standard place embedding [1] (about $+28$ to $+40$ macro-averaged F1), and most of the gain comes from the per-visit context. We then train one multi-task model for both tasks. At every state it beats a dedicated category model (about $+4.7$ to $+7.7$ macro-F1), and on region it beats the dedicated model where the region space is large and is statistically non-inferior within a two-point margin (TOST) where it is small. The spatial gain grows with the number of regions. On a non-U.S. city (Istanbul) the result holds: it beats on category and stays within two points on region.**

***Index Terms*—next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks, mobility data**

I. Introduction

Location-based social networks (LBSNs), such as Gowalla [2], let people share the places they visit. Each check-in is a small record of human movement, and together these records show how people move through a city. A natural and useful question is what a person will do next, because a service that can anticipate the next move can prepare ahead of time instead of reacting after the fact. For a mobility-aware service this can mean staging the right content in the area a user is heading to, or planning capacity there before demand arrives, and the same signal also helps recommendation and urban analysis.

Predicting the exact next place, a single point of interest (POI), is hard, and it is often more than a service needs. Two coarser questions are easier to answer and usually enough: the next category, the kind of place such as food or shopping, and the next region, the part of the city a user is heading to (we use the census tract, or the *mahalle* in Istanbul). We do not predict the exact next place. The two questions pair naturally, one for intent and one for geography, and they are more learnable than a single POI while still being useful. The question we study is whether one model should learn both at once.

Sharing one representation across tasks is not free. Our earlier work [3] found that joint training can help one task while quietly hurting the other, because the shared part is pulled toward a compromise that suits neither perfectly [4]. So the useful question is not whether multi-task learning (MTL) is good in general, but where sharing helps, where it costs, and what to do about it.

We make two changes and measure each one carefully. First, we build a check-in-level representation: instead of one fixed vector per place, where two visits to the same coffee shop look identical, each check-in gets its own vector that carries its context (the time, the nearby places, and the recent trail). This extends hierarchical graph representations of places [1], and the infomax idea behind them [5], from the place down to the individual visit (Fig. 1). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared semantic trunk and a private spatial path for the region task (Fig. 2). We evaluate on two very different settings on purpose, five U.S. states of different sizes (Gowalla) and one international city (Istanbul), so we can see whether the findings hold across small and large, U.S. and non-U.S. data.

Our contributions are the following.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit. It improves next-category prediction by a large and consistent margin over a standard place embedding, about $+28$ to $+40$ points of macro-F1 (a balanced score that counts every category equally) across all five states, and a controlled test shows the gain comes from the per-visit context, not from extra supervision (Table II).
- A single model for joint next-category and next-region prediction that shares a semantic context across the two tasks while keeping a private path for the spatial one. In one forward pass it beats a dedicated category model at every state (by about $+4.7$ to $+7.7$ points of macro-F1) and, on region, beats the dedicated model where the region space is large while staying statistically non-inferior within a two-point margin (TOST, a test of equivalence) where it is small (Table III).

- An empirical account, across five states of different sizes and one international city, of how the joint gain scales with the number of regions. The category win holds everywhere, and the region result moves monotonically: from statistically non-inferior within a two-point margin (TOST) at the small region counts to beating the dedicated model at the large, with the largest state (California) showing the largest region gain (Fig. 4). The joint results on the Gowalla states are measured over five folds at a single seed.

The rest of the paper reviews background and related work (Section II), defines the problem and the two tasks (Section III), presents our method (Section IV) and experimental setup (Section V), reports results (Section VI), discusses findings and limitations (Section VII), and concludes (Section VIII).

II. Background and Related Work

A. From place embeddings to per-visit embeddings

A place embedding turns a point of interest (POI) into a vector, so that a model can reason about places by their geometry rather than by raw identifiers. Deep Graph Infomax (DGI) [5] learns such vectors by training them to tell a real network of places, linked by similarity, time, and distance, from a shuffled copy of that network. Hierarchical Graph Infomax (HGI) [1] adds geographic structure, organizing the network as place, then region, then city, so the vectors respect where places sit. Both methods emit one vector per place, so two visits to the same coffee shop, one at 8 a.m. and one at 10 p.m., look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [6] is the closest example, learning a vector per visit by hiding parts of a user's check-in sequence and learning to fill them back in. Our representation, Check2HGI, also works at the check-in level, but it gets there a different way. It keeps the hierarchical place-to-region-to-city network and trains it with the same infomax objective, now extended down to a fourth, check-in level (Fig. 1), rather than switching to a sequence model. Put another way, earlier representations changed *which* features a place vector encodes; we change *how often* the representation emits a vector, once per place or once per visit. The novelty is this specific combination, per-visit context inside a hierarchical graph-infomax representation, and we compare against CTLE directly to show that the combination, not per-visit context on its own, is the source of the gain.

B. Predicting the next category and the next region

A large body of work predicts the next place a user will visit, the exact POI, with recurrent and attention models such as ST-RNN [7], DeepMove [8], Flashback [9], STAN [10], and GETNext [11]. We deliberately predict two coarser properties of the next visit instead: its category, what kind of place it is, and its region, where it falls, at the granularity of a census tract. Coarser targets are easier to get right and still useful to act on, and they line up with two kinds of preparation a mobility-aware service can make. We set aside the exact next place. The category task appears in our own earlier line of work [3]. The field increasingly models several granularities at once, predicting category-level and region-level sequences alongside the place sequence; in that work, however, category and region are auxiliary signals that help a headline next-place task (SGRec [12], MCMG [13], HMT-GRN [14]). We study the pair as the object itself. We also note that next region over census tracts has little precedent: HMT-GRN predicts a coarse spatial unit (a geohash cell) only as an aid, so we define our region unit carefully when we state the tasks.

C. Multi-task learning for mobility

Multi-task models for mobility have paired location prediction with related signals for some time. MCARNN [15] jointly predicts activity and location, and iMTL [16] couples them in a shared, interactive framework built for sparse check-ins. The closest alternative to our setting is the cascade. CSLSL [17] predicts in a chain (when, then what, then where), with location as the headline output and category as an instrumental step along the way; CatDM [18] uses category in the same spirit, to filter candidate places. We differ in two ways. We predict category and region in parallel, as co-equal end targets in one model with a single forward pass, and we drop next-place entirely, so neither task is instrumental to a third. On the optimization side we are deliberately conservative: hard parameter sharing across tasks is standard practice [4], and our use of it is not itself the contribution. Rather than propose yet another gradient balancer (PCGrad [19], GradNorm [20], Nash-MTL [21]), we ask where sharing helps and where it costs and give a clear, measured account, consistent with reports that such balancers rarely beat a well-tuned fixed weighting at two tasks [22], [23]. The novelty here is empirical, not architectural.

III. Problem and Tasks

We work with check-in sequences. For each user we order their check-ins in time and form short windows of nine consecutive visits. From each window we predict two properties of the next visit. The first is its *category*, the kind of place, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. The second is its *region*, the census tract the place falls in (the *mahalle* for Istanbul). We do not predict the exact next place; these two coarser properties are easier to learn and, for most uses, enough. Category is a small, fixed label set. Region is a large one, ranging from about five hundred classes (Istanbul) to about eight thousand five hundred (California), so the spatial task is a fine-grained choice over many candidate regions, while the semantic task is a choice among a handful of kinds of place.

The motivation is practical, and the two predictions line up with two kinds of preparation a mobility-aware service makes. The category says what kind of place comes next, which hints at the content or service a user will want; the region says where it is, which says where to get ready. With both in hand a service can act ahead of regional demand, for

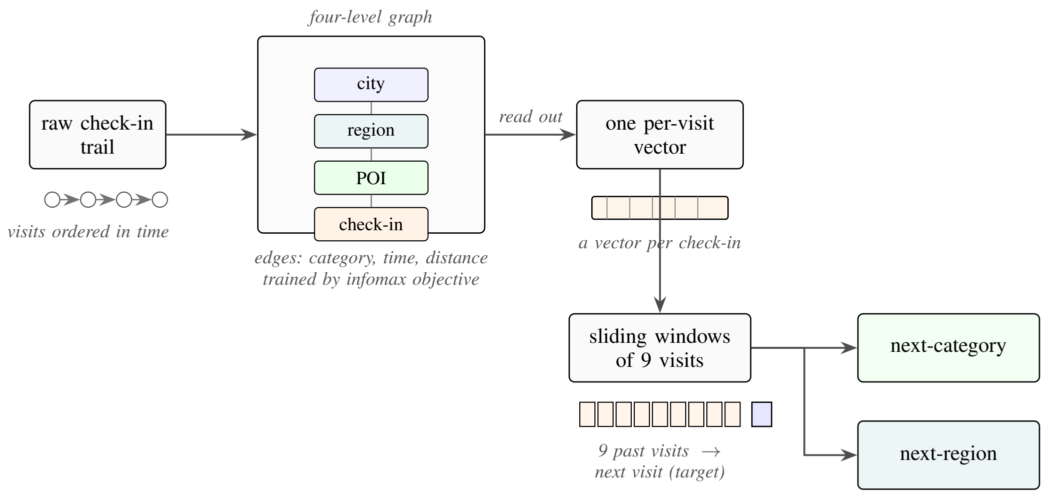

Fig. 1. From a user's check-in trail to two predictions. Each visit is placed in a four-level graph (check-in, POI, region, city) and described in its own context; overlapping nine-visit windows then feed the model, which outputs the next category and the next region.

example by staging content in the region a user is heading to, or by planning capacity there before the load arrives. These examples are sized to what we actually measure: a census tract is a neighborhood, not a radio cell, and even the ten most likely tracts are far too coarse to drive cell association or handover, so we keep the motivation at the level of regional demand and load anticipation, content staging, and capacity planning. With that framing in place, we study how well one model can anticipate both the category and the region of the next visit. We do not build or evaluate such a service, or any network system, here; the application is the reason the predictions matter, not a result we claim.

IV. Method

A. The check-in-level representation

We want each visit, not each place, to carry its own vector. A standard place embedding gives every point of interest (POI) one fixed vector, so two visits to the same café look identical even when they happen at different times of day, in different company, or after different trips. We instead describe each check-in in its own context. Figure 1 shows the full data flow, from the raw check-in trail to the two outputs.

We build a graph with four levels: the check-in, the place it records, the region (a census tract) the place sits in, and the city. Edges tie these levels together and tie check-ins to one another, encoding category similarity (visits to similar kinds of place), time (visits close in time), and distance (visits close in space). This keeps the place-to-region-to-city geography of a hierarchical graph and adds the check-in as a new bottom level. We train the graph with an infomax objective, which contrasts real neighborhoods against shuffled ones, so the vectors learn to tell a true local structure from a random one [1], [5]. The objective uses no task label: it never sees the next category or the next region. From the trained graph we read out one vector per check-in, and a model then sees a sequence of these per-visit vectors rather than a sequence of repeated per-place ones.

The pieces here are individually standard. Per-visit context, hierarchical place graphs, and the infomax objective all exist in prior work. Our one honest novelty is the combination: emitting a vector per check-in from inside a hierarchical-graph-infomax representation, a combination that earlier contextual embeddings built from sequence models do not provide, and we show later that this combination, not extra supervision, is what makes the category task far easier to learn.

B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts two properties of the next visit at once: its category and its region. Figure 2 shows the design. Two input streams pass through private per-task encoders into a shared trunk, where the two streams exchange information (a cross-attention layer lets each stream read the other's features), and the trunk feeds two outputs, one for the category and one for the region. The region output keeps a private spatial path of its own, a small branch that the category task does not touch.

We want to be precise about what that private path is, because it is easy to misread. It is a task-specific branch inside the one model, not a second model, not a second representation, and not a route that swaps models per task. The whole system is a single model: one set of shared parameters, one forward pass, two predictions. This is the deployment property we care about, since a service that wants both answers runs one model once, not two models in sequence.

The training loss is a fixed-weight sum of the two outputs, and both outputs use plain unweighted cross-entropy. We avoid class-weighting on purpose. Our category metric, macro-F1, already counts every class equally, so re-weighting the loss to favor rare classes would optimize a different objective than the one we report; we keep the objective and the metric aligned and let the equal-weight sum balance the two tasks. This is ordinary hard parameter sharing [4], kept deliberately plain so that any gain comes from the shared representation and not from a tuned weighting scheme.

The cost of serving both tasks this way is small. One joint model has about five percent more parameters and compute than a single dedicated model, far less than the two separate models the usual setup would deploy, so two answers come at roughly the price of one.

V. Experimental Setup

A. Data

We evaluate on six datasets, summarized in Table I. Five come from Gowalla, a location-based social network, and cover the U.S. states of Alabama, Arizona, Florida, Texas, and California [2]; the sixth is the city of Istanbul, drawn from the Massive-STEPS collection [24], which gives us a non-U.S. check on the findings. We chose this spread on purpose. The states run from small to large, from about 114 thousand check-ins and 1,109 regions in Alabama to several million check-ins and 8,501 regions in California, so we can watch a result hold (or not) as the region space grows. Every dataset uses the same seven place categories: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. The region unit is a census tract for the Gowalla states and a mahalle (a municipal neighborhood) for Istanbul, and the number of regions ranges from about five hundred to about eight thousand five hundred. The short reading of Table I is that our datasets span a wide range of size and region count, which is exactly what lets us study how joint training behaves with scale.

B. Windows, splitting, and the integrity of the representation

Windows. For each user we sort the check-ins in time and form overlapping windows of nine visits, with the next visit as the target. Overlapping windows give the category task more examples to learn from. We then apply a simple gate that removes a small over-representation of each user's last place (because the windows overlap, many end on the same final visit); the gate de-skews that target while keeping one full-context prediction of the last place. These are valid examples, not padding.

Splitting. We split by user with stratified five-fold cross-validation (CV), so all of a user's windows fall in the same fold. Overlapping windows therefore cannot leak across the split, because a test user's visits never appear in training.

Integrity of the representation. We train the representation once on the whole dataset and then feed it to both the dedicated single-task models and the joint model, so it is worth being explicit that it carries no information about the test visits. It does not, for three reasons. First, the representation is trained without any task label: its objective only contrasts real graph neighborhoods against shuffled ones, and it never sees the next-category or next-region target, so no label can travel from test to train through it. Second, the only thing it could carry from the test side is graph structure (which places sit near which, in space, time, and category), because the graph is built over all places. We measured that exposure directly: for each fold we rebuilt the representation from only that fold's training users and re-ran both tasks, comparing it against the full-corpus representation on the same folds. The effect is within fold noise and shows no systematic gain from the wider exposure: on region the change is -0.33 points at Alabama and -0.12 at Florida, and on category, measured on the visits whose places are seen in training (the large majority of test visits), it is $+0.29$ at Alabama and $+0.00$ at Florida. Third, the one component that really passes visit-to-visit information, the region-transition prior used by the spatial path, is built per fold from training data only, seed by seed; an earlier version that used the whole dataset inflated region accuracy by 13 to 27 points, so we removed it. Every learned baseline representation is pre-trained the same way, so all inputs are held to the same standard. The one residual we cannot fully isolate, visits to places never seen in training, is the scope of a planned training-only variant; the evidence in hand bounds any inflation to under a point on the measurable majority and finds none.

C. Metrics, superiority versus non-inferiority

For the category task we report macro-averaged F1 (macro-F1), which averages the per-class F1 so that each of the seven categories counts equally, including the rare ones; its reference point is the majority-class floor, the macro-F1 of always predicting the single most common category. For the region task we report accuracy at ten (Acc@10), the fraction of test visits whose true region appears among the model's ten highest-scoring guesses; its comparison anchor is the dedicated single-task model.

We make two kinds of comparison and test each one appropriately. Where the joint model is meant to win, we test *superiority* with a paired Wilcoxon signed-rank test over the folds. Where the joint model is meant only to stay even, we test *non-inferiority*, that it is no worse than the dedicated model by more than a two-point margin, with a two one-sided tests (TOST) procedure. We fix the two-point margin in advance and on deployment grounds, before looking at whether any claim survives it. A mobility-aware service that anticipates regional demand, stages content for the right region, or plans capacity at the level of a census tract acts on coarse regional signal, not on a single rank position, so a two-point shift in Acc@10 is below the granularity at which such a service would behave differently. The margin is also large relative to the task: with one thousand to eight thousand five hundred regions, a random top-ten guess lands the true region under one percent of the time, so two points spans a wide band of the achievable range. Finally, the equivalence is well powered. The per-fold region standard deviation is small (0.16 to 0.37 points), which gives a power near 1.0 to reject a true two-point gap, so a non-inferiority verdict reflects a real match rather than a noisy test. Which cells turn out to be a beat and which a match is a result, reported in Section VI, not here.

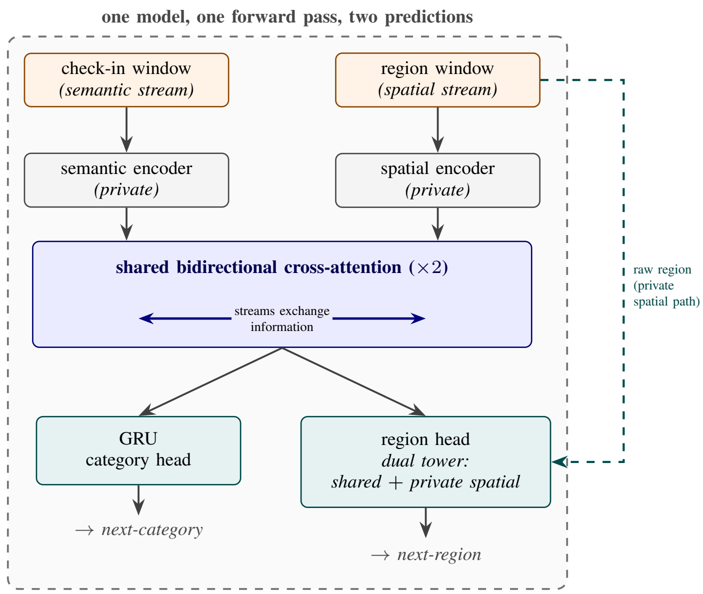

Fig. 2. The single multi-task model. Two input streams, a check-in (semantic) window and a region (spatial) window, pass through private per-task encoders into a shared bidirectional cross-attention stack (the only shared parameters). A GRU head reads the shared semantic output for next-category; a dual-tower region head combines that shared output with a private spatial tower that runs on the raw region window for next-region. One model, one forward pass, two predictions.

D. Baselines

We compare against baselines in three roles, kept separate so each answers its own question. The first role is the state of the art for each task, on our data and folds. For next-category this is POI-RGNN [25], run as a faithful author implementation, and a simple "what usually follows what" Markov baseline over the nine recent categories (Markov-9). For next-region this is HMT-GRN [14], our primary region-native comparison, predicting the region from its own end-to-end recurrent model on the same data, seed, and folds; we run the HMT-GRN-style trunk and its region-transition prior, and drop its graph module and hierarchical beam search, which exist to prune a next-place head we do not predict (so the HMT-GRN comparison is a region-native model, not a component-complete reproduction). Alongside it we run a faithful STAN [10], trained from raw with its own embeddings and its own sequence construction, and a ReHDM [26] reference reported under that method's own published protocol rather than as a matched cell. We never feed STAN our representation; a representation-fed variant would not be a faithful baseline, and we note in passing that such a variant is competitive at the smallest state, which we read as future headroom for the representation rather than a baseline. The second role is a representation control, run on Florida, to attribute the category gain to our specific design and not to contextualization in general or to extra features: CTLE [6], the closest prior contextual check-in embedding, presented fairly in both its end-to-end and frozen forms, and a feature-concat control that appends raw per-visit features to a standard place embedding under the same model. The third role is a multi-task comparator: the CSLSL cascade [17], which predicts in a chain rather than in parallel, so we can contrast our one-forward-pass joint model against the dominant alternative multi-task design at equal cost. The dedicated single-task models are the comparison anchors for the joint model throughout, and the standard place-level embedding [1] is the place-level comparison that isolates what the per-visit representation adds.

VI. Results

A. The representation makes the category task learnable

Read this as: describing each visit in its own context, rather than each place once, is what makes the next-category task learnable. Table II compares our check-in-level representation against a standard place embedding (HGI [1]) on next-category macro-F1 (the F1 score averaged equally over the seven categories, so rare ones still count), both fed to the same dedicated model under the windowing used throughout the paper. The check-in-level representation wins by a wide and consistent margin at every state: $+29.31$ at Alabama, $+27.63$ at Arizona, $+39.62$ at Florida, $+37.95$ at California, and $+37.47$ at Texas. The gains fall in two bands, about $+28$ to $+29$ points at the two small states and about $+37$ to $+40$ at the three large ones. In absolute terms the place embedding recovers only about half of what the per-visit representation reaches. A controlled test attributes most of that gain to the per-visit context itself: replacing each visit's vector with a per-place-averaged version of the same representation gives back only part of the margin, leaving roughly 64 to 90 percent of the gain (state-dependent) to the context each visit carries, not to extra training signal.

Figure 3 shows why the representation helps this task in particular. By category, the per-visit vectors are cleanly grouped.

TABLE I

DATASET STATISTICS, ORDERED BY NUMBER OF REGIONS. FIVE GOWALLA STATES AND ONE INTERNATIONAL CITY (ISTANBUL, FROM MASSIVE-STEPS) SPAN SMALL TO LARGE REGION SPACES, CHOSEN ON PURPOSE TO TEST WHETHER THE FINDINGS HOLD ACROSS SCALES AND SETTINGS. EVERY DATASET USES THE SAME SEVEN GOWALLA ROOT CATEGORIES (COMMUNITY, ENTERTAINMENT, FOOD, NIGHTLIFE, OUTDOORS, SHOPPING, TRAVEL). WINDOWS ARE THE OVERLAPPING NINE-VISIT INPUT PLUS THE NEXT VISIT AS TARGET, UNDER THE OVERLAPPING-WINDOW PROTOCOL USED THROUGHOUT; MAXIMUM AND AVERAGE SEQUENCE LENGTH ARE PER-USER CHECK-IN COUNTS. DENSITY IS THE CHECK-IN-MATRIX FILL, $100 \times$ CHECK-INS/(USERS $\times$ POIS), THE STANDARD NEXT-POI SPARSITY MEASURE.

Dataset	**Source**	**Check-ins**	**Users**	**POIs**	**Regions**	**Windows**	**Max len**	**Avg len**	**Density (%)**
Istanbul†	Massive-STEPS	462,615	23,694	29,816	520	270,217	817	19.5	0.065
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051

Region unit: a census tract (a mahalle for Istanbul). † The Istanbul row uses the mahalle-level data after dropping null-coordinate visits; the raw corpus is 544,471 check-ins over 23,700 users.

Their silhouette score by category, a cluster-separation measure from -1 to 1 where higher means tighter and better-separated groups, is about 0.56, against about 0.00 for the place embedding; and the share of a vector's nearest neighbors that carry its own category is about 0.98, against about 0.78. The same geometry does not separate regions, which is exactly the point: the representation sharpens the semantic question and stays neutral on the spatial one, so its benefit is category-only and sets up the joint study in Section VI-B. A direct comparison with the closest prior contextual embedding makes the source of the gain concrete. CTLE [6], which also describes each visit but learns it with a sequence model rather than a hierarchical graph, reaches only 29.69 to 33.45 macro-F1 at Florida across its trained variants, far below our 75.15. A feature-concat control, the place embedding joined with raw per-visit features and fed to the same model, does not close the gap either, so the category gain comes from the hierarchical per-visit representation, not from contextualization alone and not from feature injection.

B. *One model, two tasks*

Read this as: one model, in a single forward pass, beats a dedicated next-category model at every state and beats or matches a dedicated next-region model, and its spatial win grows as the region space grows. Table III reports the single joint model against the dedicated single-task ceilings, and Figure 4 plots the signed gains ordered by region count. On category the joint model beats the dedicated ceiling at every state, by $+4.68$ to $+7.69$ macro-F1 (smallest at Florida, largest at Alabama, and $+6.69$ at Istanbul). On region, measured by accuracy at ten (Acc@10, how often the true next region is among the model's ten highest-scored regions), the joint model beats the dedicated ceiling where the region space is large, Florida $+0.57$ (4,703 regions), Texas $+2.06$ (6,553), and California $+2.18$ (8,501), each with all five folds in the winning direction, and it stays statistically non-inferior within a two-point margin (TOST) where the region space is small, Alabama -0.18, Arizona -0.06, and Istanbul -0.52. The region gain increases with the number of regions, rising from -0.52 to $+2.18$ across the six settings, so the joint model helps the spatial task more, not less, as the space of regions grows; the state with the most regions has the largest region win, the opposite of a cost that grows with the size of the spatial problem. We order by region count because that is the axis of interest; region count and corpus size co-vary across these datasets, and Istanbul is a cross-setting point measured on an earlier representation (Section VI-C), so we read the trend across the six points rather than as a precise law.

We state the support honestly. The direction is unanimous: at every state all five folds favor the joint model on category, a one-sided paired Wilcoxon signed-rank test at its $n{=}5$ floor ($p{=}0.031$ per state), and the sign is consistent across all six datasets (a 6 of 6 state-level sign test, one-sided $p \approx 0.016$); the region beats at the three large states are likewise 5/5 folds each. We treat the state, not the fold, as the unit of replication, since folds within a state share data and the same trained models. The three small-state region results are tested equivalences, statistically non-inferior within a two-point margin (TOST): Alabama (-0.18; 90% confidence interval -0.46 to $+0.09$), Arizona (-0.06; -0.41 to $+0.29$), and Istanbul (-0.52; -0.65 to -0.35); all three intervals sit well inside two points. These intervals are tight because they are built on the *paired* per-fold difference, whose standard deviation is only 0.16 to 0.37 points (the per-arm fold standard deviations in Table III are larger because the two models are highly correlated across folds), so the equivalence is well powered. We do not claim per-cell family-wise-corrected significance at $n{=}5$ (a Holm correction across the six cells does not clear 0.05 at this sample size), and we leave a multi-seed confirmation ($n{=}20$) to future work.

A reader used to multi-task learning [4] expects the harder task to pay for the easier one, so it is worth saying where the category win comes from. A controlled test isolates the source: we freeze the region pathway at the start of training so it cannot learn, and therefore cannot teach the category task, yet the full category lift survives at Alabama, Arizona, and Florida (within 0.3 of the joint model and far above the dedicated single-task model). We therefore read the category win as a stronger shared trunk, obtained at no extra deployment cost (one model, one forward pass), rather than the region task teaching the category one; we report this as a finding, not a hypothesis. The joint model also stands above every external baseline on both tasks at every state (Table III): the primary region-native comparison, HMT-GRN [14] (for example California 49.61 versus our 65.66), a from-raw STAN [10] and a ReHDM reference [26] on region, and POI-RGNN [25] and a Markov floor on category, all trail it, and every region model clears a simple Markov transition floor by a wide margin. Against the closest published multi-task alternative, a cascade that predicts in a chain (when, then what, then where) [17], our parallel model reaches the same combined accuracy at the same parameter and compute cost (the difference on the joint score is about zero). We read this as a defense of the parallel design, not a win over the cascade: the chained alternative does not match our gains, so the simpler parallel structure loses nothing. The Gowalla cells are seed-0 results over five folds (n=5, provisional); the Istanbul cells average four seeds (n=20).

C. *External validity (Istanbul)*

Read this as: the finding holds on a non-U.S. city, read as a lift over the dedicated ceiling rather than as an absolute score. On Istanbul the joint model beats the dedicated category ceiling by $+6.69$ macro-F1 (53.20 to 59.89) and matches the dedicated region ceiling, statistically non-inferior within a two-point margin (TOST), at -0.52 Acc@10 (74.80 to 74.28). The category headline is the four-seed mean against a single-seed ceiling; the clean seed-0 fold-paired gain is $+6.5$ with all five folds positive, so the result does not rest on the seed averaging. We report Istanbul as a lift, not as an absolute Acc@10, because its region count (520 mahalle) differs from the Gowalla states, and we note that it is measured on the earlier graph-convolution representation (the overlapping-window representation was not rebuilt for it), so it is a cross-setting check rather than a like-for-like one. The external baselines line up the same way as on the Gowalla states: on category the joint model is far above a nine-step Markov floor (24.55) and POI-RGNN [25] (30.12), and on region it is above a faithful STAN [10] (61.86) and HMT-GRN [14] (60.4). The picture from the U.S. states repeats on a different continent and a different region unit: the joint model wins category and stays within two points on region.

VII. Discussion and Limitations

The clearest reading of our results is a design one: one model is enough. A single model, built from a shared trunk that carries the semantic context and a private spatial path that protects the region signal, does both jobs at once. The shared context lifts the next-category task sharply over a dedicated model at every state, and the private path keeps the next-region task competitive: statistically non-inferior within a two-point margin (TOST) where the region space is small, and a clear superiority (paired Wilcoxon) where it is large. The region gain is not flat. It rises with the number of regions (Figure 4), so the spatial benefit of sharing grows as the space of regions grows rather than fading, and the joint model meets or beats the dedicated single-task model on both tasks at once (Table III).

TABLE II

The check-in-level representation makes next-category prediction far more learnable than a standard place embedding. A vector per visit, not a vector per place, is what drives the category gain. Values are next-category macro-F1 (mean and fold standard deviation) under a matched single-task head; the check-in-level column is the dedicated category ceiling of Table III.

State	Check-in level	Place level	Δ
AL	**55.87** $\pm$2.4	26.56 $\pm$1.0	+29.31
AZ	**57.13** $\pm$1.3	29.50 $\pm$1.0	+27.63
FL	**75.15** $\pm$0.5	35.53 $\pm$0.3	+39.62
TX	**69.95** $\pm$0.2	32.48 $\pm$0.6	+37.47
CA	**70.26** $\pm$0.3	32.31 $\pm$0.4	+37.95

A controlled test attributes most of the gain to the per-visit context: a per-place-averaged version of the same representation recovers only part of it, leaving roughly 64 to 90 percent of the gain (state-dependent) to per-visit context. Istanbul is omitted because no place-level embedding was built for it; its result appears in Table III.

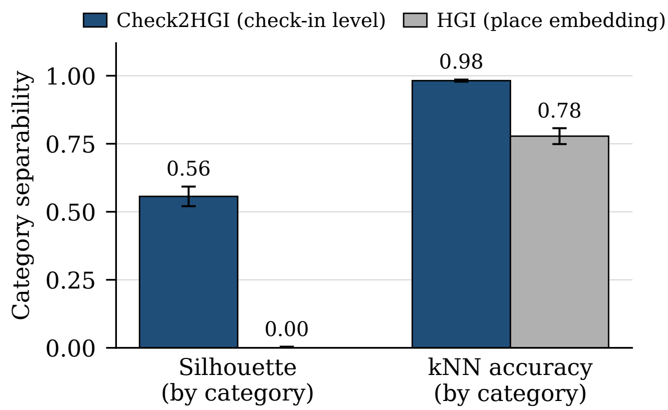

Fig. 3. Category separability of the two representations. The check-in-level representation cleanly separates the seven categories where the place embedding does not, which is exactly why it helps the category task and is neutral on region.

For a mobility-aware service this is a practical hint: one model, one forward pass, can anticipate both what a user will do next and where, without paying for a second model and without trading one task off against the other.

A short usage sketch makes the motivation concrete, and it is only motivation, since we measure no service here. A service could read the model's top-ranked next regions as an anticipatory set, a small list of areas worth preparing in advance, and read the next category as a hint at what to prepare in them. Turning that sketch into a measured result, with its own service baselines and costs, is future work, not a claim we make.

We are honest about three limits. First, the joint numbers come from a single seed over five folds, so they are provisional; a multi-seed run is the obvious next step, and it is the precondition for any per-cell significance claim stronger than the ones we report. At the smallest region counts the region result is a match, statistically non-inferior within a two-point margin (TOST), not a beat. Second, our representation is trained once over all places, so one slice of its behavior, visits to places never seen during training, is the single effect we

TABLE III

One model, two tasks. A single joint model beats the dedicated next-category ceiling at every state, and on next-region it beats the dedicated ceiling where the region space is large (FL, TX, CA) and is statistically non-inferior within a two-point margin (TOST) where it is small (AL, AZ, Istanbul). Category is macro-F1, region is Acc@10; our cells show the mean and standard deviation over five folds. Bold marks a win over the dedicated ceiling; ↑ a beat (paired Wilcoxon), ≈ a non-inferior match (TOST). States are ordered by region count.

		Next-category (macro-F1)				Next-region (Acc@10)				
State	**Regions**	**Markov**	**POI-RGNN**	**Dedicated**	**Joint (ours)**	**HMT-GRN§**	**ReHDM**	**STAN‡**	**Dedicated**	**Joint (ours)**
Istanbul	520	24.55	30.12	53.20 ±0.7	**59.89** ±0.6	60.4	–	61.86	74.80 ±0.6	74.28 ±0.7≈
AL	1,109	20.50	23.80	55.87 ±2.4	**63.56** ±2.0	57.05	66.06	60.72	69.99 ±3.2	69.81 ±3.4≈
AZ	1,547	23.92	27.64	57.13 ±1.3	**63.39** ±1.3	43.70	54.65	49.86	59.40 ±1.9	59.34 ±1.8≈
FL	4,703	29.74	34.49	75.15 ±0.5	**79.82** ±0.5	63.74	65.68	–‡	76.71 ±1.0	**77.28** ±0.8↑
TX	6,553	28.67	33.03	69.95 ±0.2	**77.51** ±0.1	53.85	–†	–†	64.96 ±0.5	**67.02** ±0.5↑
CA	8,501	27.58	31.78	70.26 ±0.3	**77.33** ±0.2	49.61	–†	–†	63.48 ±0.3	**65.66** ±0.3↑

Region externals by comparability: HMT-GRN§ (region-native, on our footing) is the primary comparison; STAN (faithful, from raw) and ReHDM (its own protocol) are secondary references. The joint model is above all three at every state, and every region model clears a Markov-1 floor by a wide margin. †Faithful STAN and ReHDM are infeasible at the CA/TX region scale; STAN reports AL, AZ, and Istanbul (FL not yet available). §HMT-GRN-style: its own recurrent trunk and a train-only region-transition prior, with the graph module and hierarchical search dropped. Baseline cells are 5-fold means. Gowalla joint-model cells are n=5 (seed 0) provisional; Istanbul cells are four-seed means (n=20) against a single-seed ceiling. A cascade comparator (CSLSL) ties our parallel model at equal cost (see text).

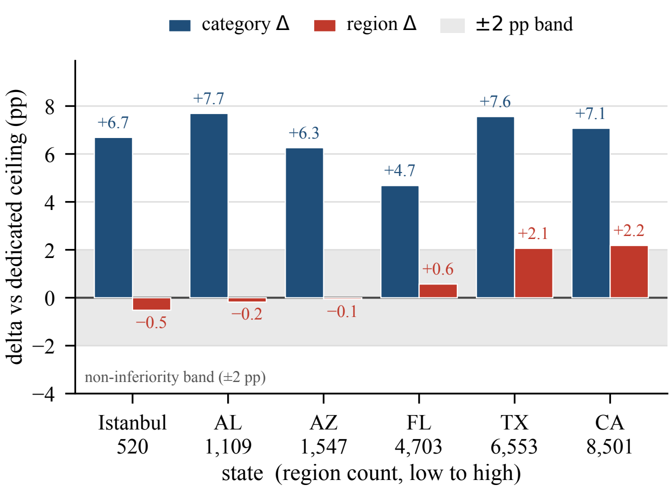

Fig. 4. Signed category and region gains of the single joint model over the dedicated single-task models, across states ordered by number of regions. The category gain (macro-F1 points) is positive at every state; the region gain (Acc@10 points) rises with region count, from within a two-point margin at the small states (AL, AZ, Istanbul) to a clear gain at the large ones (FL, TX, CA), where the largest state (CA, 8,501 regions) has the largest win. The two metrics are on different scales and are not directly comparable; the two-point band applies to region. Gowalla cells are n=5 (seed 0) provisional; Istanbul is four-seed (n=20).

cannot fully isolate; a planned variant that trains on each fold's training places only is meant to close that gap. Third, although a mobility-aware service is our motivation throughout, we do not build or evaluate one; tying these predictions to a concrete service is the natural next study.

VIII. Conclusion

We asked whether one model can predict both the category and the region of a user's next visit. Two findings answer it. First, a check-in-level representation, where each visit gets its own vector instead of one fixed vector per place, makes the next-category task far more learnable than the standard place-level representation does. Second, on top of that representation a single multi-task model beats a dedicated category model at every state, and on region it beats the dedicated model where the region space is large while staying statistically non-inferior within a two-point margin (TOST) where it is small. What we take away is not only a model but a clear reading: one model that shares a semantic context across both tasks while keeping a private spatial path for region can anticipate both what a user will do and where, and it does so without giving up the spatial task, and the spatial gain grows with the number of regions.

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